

Participatory Design

An introduction

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The aim of this book is to provide a current account of the commitments and contributions of research and practice in the Participatory Design of information technologies. An overview of the central concepts that have defined and shaped the field is provided as an introduction to the more detailed focus of later chapters. The target audience is identified and the structure of the book explained. A short description of each chapter highlights its particular contributions as well as the associated challenges facing designers and researchers engaged in participatory approaches. The chapter concludes with some guidance and recommendations for further reading.

An introduction to Participatory Design is followed by explanations of how practitioners and researchers in the field understand participation and practice and how design is approached as a process driven by social interaction and engagement. The structure of the book is described, individual chapters introduced and further relevant publications listed.

Essentially this chapter introduces, motivates and grounds the book and the chapters that follow. It provides basic definitions of the core concepts of Participatory Design and explains both their origins and ongoing relations to the motivations and commitments of researchers and practitioners who use participatory approaches in their work. The chapter provides the foundation to account for the structure of the book: one section focusing on some of the different perspectives in the field and their particular contributions and challenges and another section that presents case studies of three outstanding applications of Participatory Design.

If we are to design the futures we wish to live, then we need those whose futures they will be to actively participate in their design. This is why it is so important that Participatory Design keeps developing the design processes, tools, techniques and methods needed to enable full and active participation in all kinds of design activities.

Introduction

The beginnings of Participatory Design lie among the various social, political and civil rights movements of the 1960s and 70s, when people in many Western societies demanded an increased say in decision-making about different aspects of their lives and were prepared to participate in

collective action around shared interests and values. Some designers and design researchers responded to these events by investigating how they might relate to their own practices. For example, the theme for the 1971 conference of the Design Research Society, held in Manchester, was *Design Participation* (Cross 1972) and, around the same time, some architects and town planners began to seek ways to involve people in the design of various aspects of their everyday built environments (e.g. Sanoff 1978). The Participatory Design of information technology was pioneered in Europe and especially in Scandinavia as part of what became known as the workplace democracy movement during the 1970s (e.g. Nygaard and Bergo 1975; Sandberg 1979; Bjerknes et al. 1983). In particular it was a response to the transformation of workplaces driven by the introduction of computers. The aim was to provide people with better tools for doing their jobs, eventually enabling them to extend their skills while automating the tedious and repetitive parts of work. It is the motivation behind this aim and the context from which it emerged that has been so important to the development of Participatory Design.

The focus of this book is the tradition of Participatory Design that originated in Scandinavia and has concentrated on the design of information and communications technologies and technology-enabled systems (we will refer to these in the remainder of this book more simply as *information technologies*). At the heart of this tradition is an unshakable commitment to ensuring that those who will use information technologies play a critical role in their design. As such, the Participatory Design tradition is defined by a perspective that always looks forward to the shaping of future situations. A diverse collection of Participatory Design principles and practices has been developed, driven by ongoing efforts to deepen our understanding of how collaborative design processes can enable the participation of those who will, in the future, be affected by their results. Increasingly, designers committed to participatory approaches have extended their concerns to involving people in the design of the tools, environments, businesses and social institutions in which these technologies are embedded.

While Participatory Design has matured as a research discipline and field of design practice, it has been some years since a comprehensive volume of writings on the field has been published. In that time many of its insights, tools and techniques have become part of other design fields. Other designers and researchers have begun to use participatory approaches in their own work and new generations of both have sought to contribute to and extend the project that unites the different expressions of Participatory Design: directly involving people in the co-design of the artefacts, processes and environments that shape their lives.

The aim of this book is to provide a current account of the commitments and contributions of Participatory Design. In essence, Participatory Design can be defined as

a process of investigating, understanding, reflecting upon, establishing, developing, and supporting mutual learning between multiple participants in collective ‘reflection-in-action’. The participants typically undertake the two principal roles of users and designers where the designers strive to learn the realities of the users’ situation while the users strive to articulate their desired aims and learn appropriate technological means to obtain them.

These two principal roles reflect two fundamental aspects of Participatory Design. The first is that it seeks to enable those who will use the technology to have a voice in its design, without needing to speak the language of professional technology design. This is achieved through interactions with prototypes, mock-ups and other tools that can represent developing systems and future practices. The second is that people who are not professional technology designers may not be able to define what they want from a design process, without knowing what is possible. A

process of mutual learning for both designers and users can inform all participants' capacities to envisage future technologies and the practices in which they can be embedded.

By 'designers' we mean, at least initially, those participants who are professionally 'responsible for the information technology design project, underscoring the importance of the element of design in such projects and, in turn, the analogy to the function of the architect in construction' (Bødker et al. 2011, p. 117). By 'users' we refer to those participants who will interact with the information technologies being designed. But while this identification of the two principal roles of designers and users is useful in any simple definition of Participatory Design, in practice the roles have at times been contested and generally used with some unease. Throughout this book these roles will be challenged and elaborated to include much more than just one group's *design* of technology and another group's *use* of it. This is not just because changing technologies have continued to blur both the separation of designers and users as well as much of the gap between design and use. It is also because in Participatory Design the relations between humans and information technologies have always differentiated (among others) the users, who apply the technology for certain purposes; the customers, who order and pay for it; employees and management, who, because Participatory Design is also a political process, are involved in different ways in the conflicts and dilemmas that accompany it; and other participants, who might have first-, second- and third-hand knowledge of use-processes that, in turn, entail consequences for the tools and techniques which are helpful and relevant to apply in bringing that knowledge to bear (ibid.).

Many of the authors of the different chapters in this book have explicitly stressed that Participatory Design is not defined by formulas, rules and strict definitions but by a commitment to core principles of participation in design. These, in turn, are informed by a rich heritage of projects, methods, tools and techniques that we can bring to bear on each specific design context in which we participate. Both the shared principles of Participatory Design and many of the ways these have been expressed and experienced have informed the structure of this book and are examined from different perspectives in its pages.

We continue in this introductory chapter by providing an overview of the central concepts of Participatory Design. We discuss the motivation behind its basic commitment to participation, what we understand by participation and practice, and how design is understood as a process driven by social interaction and engagement. From there the chapter identifies its target audiences and explains the structure of the book, paying particular attention to how its chapters articulate the contributions of Participatory Design as well as the challenges that lie ahead. For those who want to read more, the chapter closes with some suggestions for further reading and provides a comprehensive list of Participatory Design publications.

Taking a stand

Writing of the early Scandinavian Participatory Design projects of the 1970s and 1980s, Morten Kyng observed: 'As part of the transformation of the workplace working conditions for many end users have changed dramatically, and not always for the better' (1988, p. 178). Discussions about these changes, between designers, users, trade unions and other stakeholders, highlighted that when those who would use these new computer-based systems were not actively involved and influential in their development and use, they were unable to create visions of future working conditions and practices that would improve or even match their current ones. Participatory Design researchers and trade unions initiated a range of activities to question existing approaches to the computerisation of the workplace and to influence legislation on the changing work environment. Active involvement from those who would use these systems was the key characteristic of these newly initiated activities; they were intended to achieve their aims through

creating visions of different kinds of future workplaces and practices and to design the new computer-based systems that would shape them (ibid.). This meant that new ways of designing systems needed to be developed that, in turn, relied on finding new ways of cooperating between users and designers.

Then, as now, participatory designers viewed those who would use the new technologies as experts in their work domains while the designers acted as experts in theirs, contributing as experts and facilitators of the design process, technical consultants and often through involvement in the actual development of new systems. So Participatory Design was, in the first instance, about designing information technologies that would allow people to change and further develop their work practices to incorporate both the use of computer systems and improved working conditions. Its roots in the workplace democracy movement meant that those whose new work practices were being designed were centrally engaged in the process. This engagement was itself defined and enabled by the development of new design methods, tools and techniques that were intended to enable all those involved in the design process to imagine and move towards new visions of the future.

The commitments of Participatory Design to new ways of designing and to people's agency in shaping their work marked a clear move away from the formalised models of work that were, at the time, dominant in information systems design, business planning and related areas. Joan Greenbaum and Morten Kyng devoted a major part of their introduction to *Design at Work* (Greenbaum and Kyng 1991a, pp. 1–24) to critiquing the cultural, political and economic values inherent in formalised models of human activities and explicitly connected these models to the deeply rooted practices of the rationalist tradition of Western philosophy. They cited the summary provided by Winograd and Flores (1986, p. 15) of how these practices characterise situations in terms of identifiable objects with well-defined properties; find general rules that apply to situations in terms of those objects and properties; and then apply the rules logically to the situation of concern in order to draw conclusions about what should be done (Greenbaum and Kyng 1991a, p. 7). The effects include a top-down view of the organisation that formalises work into disembodied, algorithmic procedures and makes invisible the social, embodied and contingent nature of everyday work practices. Indeed, critiques of formalised representations of work are common in the early Participatory Design and related literature (e.g. Suchman 1983, 1987, 1995; Ehn and Kyng 1984; Kyng 1988, 1995; Greenbaum and Kyng 1991a; Star 1991). These critiques extended to the way such models informed developing approaches to system development contributing both to the failure of new systems and to the reduced workplace conditions that had motivated the early Participatory Design initiatives.

Traditional methods and practices are firmly rooted in engineering and natural sciences origins of computer science. They are designed for implementing clear-cut specifications (as if such things exist), but they are far removed from actual practice. In fact, traditional methods, with their emphasis on step-by-step procedures, effectively prevent creative and cooperative sparks between system designers and users.

(Greenbaum and Kyng 1991b, p. viii)

Participatory Design has always sought its theoretical and conceptual inspiration in those fields and approaches that shared both its critical stance towards rationalist accounts of human practice and its focus on the primacy of human experience, social agency and institution-shaping behaviour (e.g. Winograd and Flores 1986; Suchman 1987; Ehn 1988; Greenbaum and Kyng 1991a). Most importantly, Participatory Design has always given primacy to human action and people's rights to participate in the shaping of the worlds in which they act. In their critique of rationalist

approaches to technology design, widely read and cited within the early years of Participatory Design, Winograd and Flores wrote: ‘We encounter the deep questions of design when we recognize that in designing tools we are designing ways of being’ (1986, p. xi). Here, then, is the crux of where and why we *take a stand*. Design is, fundamentally, about designing futures for actual people. Designing technologies that people rely on in their work shapes, in essential ways, how it is possible for people to do their work and how they experience their working lives. There is an ethical stand underlying Participatory Design that recognises an accountability of design to the worlds it creates and the lives of those who inhabit them. Designing in genuine partnership with those who will use the technologies we build is our way of taking a stand about who we can be as designers and design researchers.

Much has happened and many, many things have changed since these early Participatory Design projects in Scandinavian workplaces. Many of those reading this book will never experience a workplace or even a living space that has not been deeply shaped by information technologies. These technologies have moved off the desktops and out of workplaces into our homes, our public spaces, our pockets, handbags and backpacks, our leisure time and our social relationships. Such developments have constantly posed wonderfully interesting but also major challenges to those working within Participatory Design; we need to ensure that our understandings of design and our design processes, methods, techniques and tools could enable the design of new and emerging technologies for changing and expanding contexts. This book is intended to give an account of some of the ways that those working within Participatory Design have met the challenges of change and how Participatory Design has developed in the process as well as outlining the challenges ahead that current and future Participatory Design researchers need to address.

Participation

Participation is the core of Participatory Design. The fundamental developments in the field have come from efforts to investigate, understand, support and practise what can be labelled as ‘genuine’ participation in design (Bødker et al. 2004). By ‘genuine’ participation, we refer to the fundamental transcendence of the users’ role *from* being merely informants *to* being legitimate and acknowledged participants in the design process. This role is established – for example – when users are not just answering questions in an interview about their point of view or knowledge of a particular issue, but are asked to step up, take the pen in hand, stand in front of the large whiteboard together with fellow colleagues and designers, and participate in drawing and sketching how the work process unfolds as seen from their perspectives. Inviting users to such collective discussions and reflections requires a trustful and confiding relationship between all participants. Any user needs to participate willingly as a way of working both *as themselves* (respecting their individual and group’s/community’s genuine interests) and *with themselves* (being concentrated present in order to sense how they feel about an issue, being open towards reflections on their own opinions) as well as *for the task and the project* (contributing to the achievement of the shared and agreed-upon goals of the design task and design project at hand) (Storm Jensen 2002). Such participation demands acknowledgement of the users’ interests as fully legitimate elements of the design process. It includes addressing questions such as: what does participation mean in terms of actual power to make decisions; who needs to participate and how can this be managed and supported; how can the design process itself be designed so people can participate in it; and what kinds of design tools and methods do we need?

Participatory Design has had a profound influence on the current recognition and acceptance of the value of user participation in design. Improving both the quality of the product (e.g.

Grudin 1991; Grønbaek et al. 1993) and the quality of the process (e.g. Bødker et al. 2004) have always been central commitments. Involving people with everyday knowledge of a particular domain of human activity in the design of technologies for that domain offers a way to increase product and service quality because the resulting technologies work better. Using design-by-doing methods, such as mock-ups and prototypes, is another central commitment of Participatory Design because these have enabled ‘ordinary workers’ to use their practical skills when participating in the design process (e.g. Ehn 1993). Using these methods enhances participant engagement in the process and encourages more robust communication and shared understandings between designers, users and other stakeholders in design. These days, participation within information technology design is routinely advocated in many textbooks (for example, within Human–Computer Interaction) through iterative design techniques such as mock-ups and prototyping.

There are both pragmatic and political rationales for genuine participation in design. The pragmatic rationale stresses the need for users and developers to learn together about possible and useful technical solutions. So *mutual learning* and the setting up of *mutual learning processes* are defining commitments of Participatory Design. The mutual learning process typically includes ideas and visions for change as they evolve through a design project ‘that reveals goals, defines problems, and indicates solutions, with the aim of designing sustainable uses of IT based on a specific problem within the company’ (Bødker et al. 2004, p. 13).

Mutual learning is an ongoing activity throughout the Participatory Design process. Users need knowledge of potential technological options as well as of how these options can be provided. Designers are the source of this knowledge, as well as of relevant design expertise. The designers need knowledge about the users, their practices and the use situation (often the work domain) in question. The users are the source of this knowledge and relevant domain expertise. They provide experiential knowledge of their expertise, the work domain and practices. During a Participatory Design process all participants increase their knowledge and understandings: about technology for the users, about users and their practice for designers, and all participants learn more about technology design. Kensing and Blomberg summed up the pragmatic argument for full and genuine participation in design: ‘The epistemological stand of participatory design is that these types of knowledge are developed most effectively through active cooperation between workers [end-users] (and increasingly other organizational members) and designers within specific design projects’ (1998, p. 172). Mutual learning supports the design of technology based on the logic of the practice it is intended to support. This makes the solution more robust and sustainable (e.g. Bjerknes and Bratteteig 1988).

The political rationale for genuine participation in design reflects a commitment to ensuring that the voices of marginalised groups and communities are heard in decision-making processes that will affect them. The motivation was and remains democratic and emancipatory: participation in Participatory Design happens, and needs to happen, because those who are to be affected by the changes resulting from implementing information and communications technologies should, as a basic human right, have the opportunity to influence the design of those technologies and the practices that involve their use. This emancipatory commitment will be explained and explored further all through this book, particularly in [Chapter 2](#) on the heritage of Participatory Design and [Chapter 4](#) on ethics and design. Expressed simply, if we are to be able to design the futures we wish to live, then we need those whose futures they will be to actively participate in their design. This is why it is so important that Participatory Design keeps developing the design processes, tools and methods needed to enable full and active participation in design activities. As we work together to design solutions to all kinds of complex issues that confront us, not only those involving new information technologies, we need to ensure

not just that we have the best processes, tools and methods available to work with but that these can enable the successful development of inclusive and democratic design solutions.

Practice

Through practice we produce the world, both the world of objects and our knowledge about this world. Practice is both action and reflection. But practice is also a social activity; it is produced in cooperation with others. However, this production of the world and our understanding of it takes place in an already existing world. The world is also a product of former practice. Hence, as part of practice, knowledge has to be understood socially – as producing or reproducing social processes and structures as well as being the product of them.

(Ehn 1993, p. 63)

Participatory Design is driven by a consistent socio-technical approach that appreciates the context in which the technology will be used and the processes and practices within that context. The term *practice* is used to refer to what people really do. Its use emphasises that what people really do is often very different from the way their activities might be described by others or depicted in, for example, work flow diagrams and similar kinds of representations of work, or presented as what people should do, ought to do or might ideally do ‘in theory’ (Schmidt et al. 2007). Understanding practice is fundamental to Participatory Design for many reasons, most importantly because so much of what we do is guided by the recognition that designing the technologies people use in their everyday activities shapes, in crucial ways, how those activities might be done. Suchman and Trigg summarised: ‘Effective design involves a co-evolution of artefacts with practice’ (1991, p. 72). They explained:

Because of the intimate relation between work and technology, the development of the artefacts with which people work and the development of their work practices go hand in hand. Available technologies afford certain resources and constraints on how the work gets done and people’s ways of working give the technologies their shape and significance.

(Suchman and Trigg 1991, p. 65)

Rich accounts of practice and the understandings they enabled were considered early in the development of Participatory Design as alternatives to the formal diagrams and abstracted work procedures that had guided so much early technology design. At the time this commitment to the empirical study of practice, in order to understand work and inform the design of technologies that could be used in that work, separated Participatory Design approaches from other technology and work design traditions. The focus on practice recognises the role of everyday practical action in shaping the worlds we live in. That is, practice plays a central epistemological role in Participatory Design that complements its rejection of formalised models of both work and design along with their focus on individual work tasks. Most importantly, practice is, by definition, a social activity; ‘it is the community, rather than the individual, that defines what a given domain of work is and what it means to accomplish it successfully’ (ibid., p. 73). Participatory Design has developed a range of design techniques, such as various kinds of design and future workshops, to support participants working together to develop detailed accounts and other representations of both current and desired future practices. These are explored further in [Chapter 7](#) of this book.

The work by Lucy Suchman and her Work Practice and Technology Group at Xerox Palo Alto Research Center (PARC) was a major inspiration for the importance of understanding practice in both Participatory Design and Computer Supported Cooperative Work (CSCW).

There has always been a close relationship and much crossover between researchers and practitioners in Participatory Design and CSCW. Appreciating the realities of a work-oriented organisational context implies viewing work as a socially organised and situated activity, where the actual behaviour, because of its situatedness, cannot be fully described by plans (and other formal descriptions) that guide and/or help coordinate a practice (Suchman 1987, 2007). Both Participatory Design and CSCW recognise that human activities are carried out in cooperation with others and so new technologies need to be designed to support cooperation. They also share a major reliance on applying ethnographic methods to support the development of a thorough understanding of current practices, within the particular design domain, as a basis for the specification of new technologies (e.g. Blomberg et al. 1993). As described in [Chapter 5](#), such methods are an important means within Participatory Design to develop experiential understandings of actual practice and have been used extensively from its earliest days (e.g. Bjerknes and Bratteteig 1995).

Practice evolves over time as practitioners actively engage in reviewing and modifying their practice in response to all kinds of changes and developments, not just those resulting from the introduction of new technologies. But the introduction of new technologies will always change the practice they are part of, in ways that were predicted and in ways that were not, no matter how well the original domain was understood. Planned or anticipated changes may or may not happen while other changes can emerge spontaneously or as a result of the new possibilities experienced from using the technology (e.g. Orlikowski and Hofman 1997). There are variations in the same basic practice across different local settings and there are deviations from them. Specific practices and how they are defined can be questioned, evaluated, debated, undermined, and challenged, but also defended, promoted, adopted, applied, enforced, and upheld by instruction, training, and sanction (Schmidt et al. 2007).

Understandings of practice developed and used in Participatory Design include what is routinely done in some kind of human activity and how the activity is actually accomplished by the actions of those participating in it. It is the latter that provides the major link into the design of future practices and the technologies that will be part of them. The aim is to design technology not just to support cooperation between those using it but also to enable people to take advantage of the new technologies, to (re-)configure and appropriate them and to redesign their practice in positive ways over time. As we will see in the following section and in greater detail in [Chapters 6](#) and [7](#), Participatory Design has also developed a range of methods, tools and techniques to assist future users, designers and other stakeholders envisage, experience and reflect on the future practices that new technologies might enable.

Design

Participatory Design is driven by social interaction as users and designers learn together to create, develop, express and evaluate their ideas and visions. Shared experimentation and reflection are essential parts of the design process. Pelle Ehn (1993) wrote: ‘the origin of design is in involved practical use and understanding, not detached reflection, and design is seen as an interaction between understanding and creativity’ (p. 62). So Participatory Design has much in common with Schön’s (1983) articulation of design as ‘reflection-in-action’, a process engaged in by an experienced and reflective design-practitioner. But while Schön takes the perspective of the individual designer, the direct involvement of future users in a Participatory Design process involves collective ‘reflection-in-action’ between multiple participants. Within Participatory Design, what is being designed is both the technological *product* or artefact and the *process* that enables different participants to engage in designing this product. Indeed, Participatory Design has

a strong focus on the ‘how’ of designing, i.e. a focus on the process of designing and the particular participatory practices that different processes can enable (see [Chapter 9](#) for an elaborated practical example).

The development and use of a variety of design methods, tools and techniques to support collective ‘reflection-in-action’ have historically formed a substantial part of the Participatory Design agenda since the pioneering projects in the 1970s, and these are among the major contributions Participatory Design has made. Workshops with designers and users using techniques such as mock-ups, scenarios, prototypes and various types of design games have been central to Participatory Design since it began. These enable ongoing design experiments to visualise, simulate and experiment with selected elements of envisioned technologies, in use, prior to their development and implementation. They enable the participants in the design process to propose, represent, interrogate and reflect on different aspects of the developing design continually throughout that process. These shared cycles of design experiment and evaluation are the essence of the Participatory *Design* process. The ideas and visions for new technologies are based on understandings of the actual use context, but the implementation and use of new technologies changes the way activities are done. This fosters new possibilities in two ways: by providing a (new) starting point for the current design situation and new experiences and understandings of how different technological options might support new practices in the future.

Initially, Participatory Design mostly addressed small-scale and custom-made systems (Clement and van den Besselaar 1993; Oostveen and van den Besselaar 2004) or the initial parts of larger-scale information technology development followed by a conventional contractual bid (Kensing 2000; Bødker et al. 2004). But organisations now are increasingly buying generic software (Bansler and Havn 1994) as off-the-shelf systems or development frameworks offering standardised, domain-specific ‘building blocks’ that can be rapidly configured to support individual requirements. At the same time the growing use of technologies outside the work context and the availability and wide uptake of various social technologies have massively extended the domains where Participatory Design is being used.

New technologies and domains of use have enabled the expansion of Participatory Design to embrace implementation and real use of the technologies as well as ongoing redesign and reconfiguration (Büscher et al. 2004; Hansen et al. 2006; Simonsen and Hertzum 2010, 2012). Using actual technologies during the Participatory Design process offers valuable opportunities. Simulated use of technologies through mock-ups, prototypes and design games can investigate and evaluate anticipated change, but using the actual technologies means that unanticipated changes can also emerge, including inappropriate, undesirable or disastrous side effects as well as unforeseen opportunities and unintended advantageous and desirable possibilities.

The recognition that design is completed in use was shared by many of those involved in Participatory Design and has become more central in recent years (e.g. Balka and Kahnemouli 2004). During the 1990s it manifested as a marked interest in the tailorability of systems (e.g. Trigg and Bødker 1994; Robertson 1998) so users could adapt them to suit their needs after implementation. Tailorability extended to configurability with the increasing availability of off-the-shelf products and the rise of domestic, mobile and embedded technologies. Configurability has more to do with assembling and recombining components into useful devices and services (Balka et al. 2005). But the Participatory Design literature increasingly includes studies of technology use and reconfigurations of technologies in support of new and unanticipated use. This can be generalised as a move to ‘designing for design after design’, which deliberately defers some aspects of design until after a design project is completed as a way to explicitly support the potential for redesign for unanticipated use or other unanticipated change (A.Telier 2011).

[Chapter 3](#) of this book is devoted to a detailed account of how those using participatory approaches have conceptualised and practised design in their work.

Target audience, aims and structure

This international handbook is written by well-known and experienced researchers and practitioners of Participatory Design who are all deeply embedded in the field and familiar with its development, especially as it has unfolded within the Participatory Design conferences.

Our aims for this book are that it will become the major text for Participatory Design courses at colleges and universities worldwide and that it will endure and remain a relevant reference for the next decade and beyond. Our goal has been to produce a state-of-the-art reference handbook that provides an introduction to, and detailed exploration of, core areas of Participatory Design including thorough literature reviews of both central research contributions and current and future research areas and application domains.

The book's target audience is researchers and practitioners who are already working within Participatory Design, those who are interested in doing so, or those who wish to obtain an authoritative overview of the field and its history. In particular it is intended for:

- *Young researchers* looking for a scholarly handbook on Participatory Design. The book provides emerging researchers, for example graduate students and PhD candidates, with a thorough overview of Participatory Design and a familiarity with the aims and interests of the Participatory Design community. It is also intended to provide insights into potential challenging research areas that these readers might wish to address in their future careers. The contributors to the book have each provided personal statements about why they are so engaged in Participatory Design. These statements reflect the very special qualities of this research area.
- *Senior researchers* who may enjoy the book as the new major reference handbook for Participatory Design. It provides updated overviews of the major areas of the field that may be secondary yet important supplements to existing research interests and expertise. The book is designed to make available the basic literature for teaching advanced bachelor, graduate or doctoral courses. It includes an introduction to Participatory Design, accounts of what the field can offer theoretically, methodological approaches, techniques and tools supporting specific activities, and detailed examples of practical application.
- *Practitioners* who use, or wish to use, participatory approaches in their design practice. As we have discussed in this chapter, a defining characteristic of Participatory Design research is the consistent and systematic engagement with practice. Indeed, very early on Participatory Design was described as research in 'Systems design for, with, and by the users' (Briefs et al. 1983) with action research in collaboration with industry as the preferred research approach. Practitioners might especially value the impressive collection of practical applications of Participatory Design given in [Section II](#). These could also be used as preliminary readings to entice practitioners into the theoretical and methodological material and issues presented in [Section I](#).

Other very popular books have been published on Participatory Design. The two books published by Lawrence Erlbaum (Greenbaum and Kyng 1991a; Schuler and Namioka 1993) have been to date the key handbooks available on Participatory Design and they are widely cited. However, they are out of date in some important areas and do not address the major developments in the field over the past 20 years. This has meant that readings in Participatory Design

have needed to be heavily supplemented with various newer articles and papers. This book is intended to encourage reflection on, and learning from, the considerable achievements of Participatory Design research and practice over the past three decades. Most importantly, it is also intended to encourage those contributing to Participatory Design to consider how we might take up the challenges of changing societies, technologies and contexts for technology use today and in the future. It demonstrates why Participatory Design is such a valuable approach to design in general, describes the Participatory Design community and explains why its members have embraced Participatory Design as an important, highly relevant and most rewarding area for research and practice.

The book is structured into two sections.

- *Section I* includes seven chapters each representing a distinct and influential area of Participatory Design. Each chapter presents major research contributions and future research agendas and application domains based on thorough reviews of the literature from the last 30 years (see note at the end of this chapter for a list of sources). Together these seven chapters provide a thorough review of the important areas of Participatory Design research to date, while also looking forward to possible future trends, phenomena, developments, views, etc. that will challenge both the traditions and the current wisdom in the field.
- *Section II* includes three chapters demonstrating the practical, large-scale application of Participatory Design. Each chapter describes and reflects on a project that received the Participatory Design Conference's Artful Integration Award (in 2004, 2006 and 2008 respectively). This award is given to projects nominated and recognised by the Participatory Design community as outstanding achievements by a group of people who, in exceptionally creative ways, have together designed new and useful configurations of technologies and practices. While traditionally design awards have gone to individual designers or single objects or products, the Artful Integration Award emphasises the importance of collaborative participation in design, viewing good design as the effective alignment of diverse collections of people, activities and artefacts.

Section I: Participatory Design – contributions and challenges

Chapter 2 opens this section with a review of the history and heritage of Participatory Design. This history begins with the world famous projects conducted in collaboration with trade unions in Scandinavia during the 1970s and the early struggles to give a voice to those who traditionally lack power in the development process. It continues with the Participatory Design movements in the USA and Canada in the 1980s, where the society of Computer Professionals for Social Responsibility (CPSR) was formed, along with growing recognition that design and development must be situated in the real, everyday actions of people using technology. From there the chapter proceeds through the 1990s, when the Participatory Design Conferences began, to the present, when Participatory Design has become influential even in commercial large-scale information technology projects and has gained widespread academic recognition in a broad range of conferences and journals. The discussion of the heritage of Participatory Design demonstrates how difficult it is to go beyond just 'involving' users in design, to bringing about more equal power relations and genuinely fostering emancipatory participation in actual design projects.

Chapter 3 investigates and elaborates on the design concept. It provides a detailed account of how those using participatory approaches have conceptualised and practised design in their work. Relations and influences between various design traditions and Participatory Design are discussed from the birth of the design movement in 1919, the Bauhaus, to the current design

research movement. The historical account of design traditions is complemented by a discussion of the challenges that confront Participatory Design today and how these challenges might be addressed in a ‘designerly’ way. Recent research initiatives are outlined, pointing to how Participatory Design reinvents itself to face the changing political, economic, social and technological environments. The chapter finally argues that Participatory Design should shift the frame of reference from design projects to design *things* – referring to an ‘assembly’ around ‘matters of concern’, which takes place at a certain time and in a certain place. A Participatory Design thing strategy is especially relevant when aspects of democratisation are at stake.

Chapter 4 is dedicated to the ethics of Participatory Design. Participatory Design is exceptional as an international community of researchers and practitioners who expect, and are expected to, engage in the ethics of their practice and who take that engagement very seriously. The chapter explores the defining role of ethics in the development of Participatory Design. An overview of different approaches to ethics is followed by an exploration of the ethical issues of working with users. Its discussion of core ethical issues of Participatory Design is grounded in particular examples drawn from the literature where the expression of these principles has been negotiated by designers and other participants. The chapter calls for new and extended participatory methods, tools and processes that can enable and improve genuine participation, not just in the design of new technologies but in the shaping of future societies. In particular it identifies the need for new and extended Participatory Design methods and tools that can help us envision and resolve ethical issues that might arise in the use of future technologies.

Chapter 5 re-examines the relationship between ethnography and Participatory Design. It provides a much-requested invaluable update on the earlier, influential and much-cited chapter on ethnographic methods and their relation to design (Blomberg et al. 1993) which was included in *Participatory Design: Principles and Practices* (Schuler and Namioka 1993). The chapter examines the commitments and guiding principles of ethnography and Participatory Design, revisits foundational research on the application of ethnography to Participatory Design and explores how ethnography can support the important role of reflexivity in Participatory Design. Given the challenges of designing for large-scale systems, global connections and the need for the participation of people from diverse cultures and socio-economic circumstances, the chapter explains how the sensibilities of ethnography can continue to make an important contribution to the future of Participatory Design.

Chapters 6 and 7 review and discuss ways to do Participatory Design and engage with users in Participatory Design projects. Chapter 6 is devoted to methods and Chapter 7 to tools and techniques. Methods are considered as ‘recipes’ for how to do Participatory Design, and include tools and techniques as well as general guidelines and organising principles. A rich repertoire of Participatory Design tools and techniques has been developed that can be combined, adapted and extended to be used in new and existing methods and other participatory practices.

Chapter 6 explores how various fundamental commitments of Participatory Design can be translated to ways of organising Participatory Design projects and general guidelines for the design process itself. A selection of Participatory Design methods are described and examined, and inherent perspectives relevant for all Participatory Design methods are discussed in terms of having a say, mutual learning and co-realisation. This chapter addresses some of the fundamental challenges of actually doing Participatory Design projects while maintaining its defining aims and commitments. These include: the challenges of ensuring users continue to have a say in design within the current specialisation of information technology design; how new technologies imply that design increasingly continues after they are implemented; and the challenges to managing the scope and effect of user participation as the organisation of work continues to change.

[Chapter 7](#) presents a rich variety of tools and techniques that can be used in Participatory Design projects to enable various practices of participation. Using a range of examples the chapter explains how these tools and techniques are not intended to be rigidly applied, but instead used in ways that can support participants in making a variety of things, telling stories and enacting aspects of future design and, in the process, forming new, temporary communities that can enable the collective envisioning of future practices. Participatory tools and techniques avoid the formal models and abstract representations of traditional design approaches, and allow designers and users to experiment and evaluate various innovative design possibilities. In turn, this means they can be used and extended to meet the challenges of designing new technologies and practices, with different and diverse groups of users in new and changing design contexts.

[Chapter 8](#) recognises community-based Participatory Design as a distinctive field of Participatory Design research. It draws together and examines a collection of previously published Participatory Design projects that cover different notions of ‘community’ based in, for example, actual community-based organisations, loosely coupled communities such as local neighbourhood networks, and online communities facilitated by Web 2.0 technologies such as Twitter, Facebook, etc. The complexities, potentials and pitfalls of community-based Participatory Design are considered and the chapter argues for the use and further development of pluralistic approaches to employing and analysing Participatory Design practices across the diverse contexts of different communities and the issues that affect them. The authors suggest that this will require inventiveness with new modes of infrastructuring, attentiveness to new forms of politics and the exploration of new opportunities for community-based Participatory Design.

Section II: Outstanding applications of Participatory Design

[Chapter 9](#) provides one of this section’s three cases of practical large-scale applications of Participatory Design. Randy Trigg (former Participatory Design researcher from Xerox PARC) and the Global Fund for Women organisation received the first Artful Integration Award, which was given in 2004. This chapter describes a Participatory Design project that has been ongoing for more than ten years. The project has resulted in the design of a database system for a non-profit organisation that brings together fundraising and grantmaking in ways that accommodate the continually evolving work practices of the organisation. The Global Fund’s developing infrastructure weaves together Randy and his colleague Karen Ishimaru’s long-standing commitment to cooperative design practices with the Global Fund’s commitment to democratic forms of resource mobilisation. The Fund makes grants to seed, support and strengthen local women’s rights groups based outside the United States working to address human rights issues. The Global Fund has developed a unique grantmaking process based in a global network of sister organisations, and is recognised and admired for its respectful and responsive relations to its grantees. Since 1987, the Global Fund has granted \$90 million to more than 4,000 women’s organisations in 172 countries.

[Chapter 10](#) describes the second Artful Integration Award initiative: the Health Information Systems Programme (HISP), an extensive, ongoing international collaboration in health information systems for public health and higher education. HISP comprises a network of related systems development projects, as well as a dual international master’s programme in Information Systems and Public Health of the informatics and medical faculties of the University of Oslo, Norway, and the University of Eduardo Mondlane, Mozambique, with additional collaborations including the University of the Western Cape, South Africa, and the University of Dar es Salaam, Tanzania. More than two dozen international and Norwegian doctoral and master’s students participate in this programme as teachers, trainers, mentors and advisors for and with

health care personnel, as well as pursuing their own research. The open source HISP software development and implementation collaboration now extends to a growing network of countries including India, China, Malawi and Vietnam. Emerging cloud-based infrastructure means that local capacity in the new technologies needs to be developed, offering further opportunities for the use and development of participatory approaches.

Chapter 11 presents the ACTION for Health project that was initiated in 2003 and ended in 2008. The project focused on the social implications of information technology and its introduction into the health system, and aimed to fill gaps about existing knowledge of Canada's health information infrastructure. Health technology in Canada, as in many other countries, is reviewed in terms of its clinical and cost effectiveness. ACTION for Health goes beyond the limits of such traditional assessment by considering the social, ethical and legal aspects of introducing new technologies into the health sector. Through innovative Participatory Design approaches, the project enhanced health technology assessments, leading to improved policies and practices. The chapter draws attention to and explores power relations in Participatory Design projects, providing examples of how these were negotiated during the project and suggestions for those who might find themselves in similar situations.

Further reading

Today, Participatory Design is a well-established area of research and an emerging practice within many areas of information technology design. The Participatory Design research community have since 1990 met at the biennial international Participatory Design Conferences (PDCs).¹ Participatory Design is also represented in many other local and international conferences, as well as in a number of books and special journal issues.²

For those who wish to know more about Participatory Design we have in the endnotes provided a list of further readings. This includes a full list of the Participatory Design Conferences (and their themes) from 1990 until 2012. All conference proceedings from 1990 until today are available from <http://pdcproceedings.org> (1990–2002 as free downloads and from 2004 via the ACM Digital Library). The list also includes the major books on Participatory Design, other related books and the special issues of international journals that have published Participatory Design research.

Notes

¹ Below is a list of the Participatory Design Conferences (and their themes) held from the start in 1990 and until 2012:

1990, Seattle, WA, USA (Namioka and Schuler 1990)
1992, MIT, Cambridge, MA, USA (Muller et al. 1992)
1994, Chapel Hill, NC, USA (Trigg et al. 1994)
1996, Cambridge, MA, USA (Blomberg et al. 1996)
1998, Seattle, WA, USA: Broadening Participation (Chatfield et al. 1998)
2000, New York, NY, USA: Designing Digital Environments (Cherkasky et al. 2000)
2002, Malmö, Sweden: Inquiring into the Politics, Contexts and Practices of Collaborative Design Work (Binder et al. 2002)
2004, Toronto, Canada: Interweaving Media, Materials and Practices (Clement et al. 2004a, 2004b)
2006, Trento, Italy: Expanding Boundaries in Design (Jacucci et al. 2006a, 2006b)
2008, Bloomington, Indiana: Experiences and Challenges (Simonsen et al. 2008)
2010, Sydney, Australia: Participation :: The Challenge (Bødker et al. 2010)
2012, Roskilde, Denmark: Embracing New Territories of Participation (Halskov et al. 2012)

2 Books on Participatory Design:

- Bjerknes, G., P. Ehn and M. Kyng (eds) (1983) *Computers and Democracy – A Scandinavian Challenge*, Aldershot: Avebury, Gower.
- Briefs, U., C. Ciborra and I. Schneider (eds) (1983) *Systems Design for, with, and by the Users*, Amsterdam: North-Holland.
- Greenbaum, Joan and Morten Kyng (eds) (1991) *Design at Work: Cooperative Design of Computer Systems*, Chichester: Lawrence Erlbaum Associates.
- Schuler, Douglas and Aki Namioka (eds) (1993) *Participatory Design: Principles and Practices*, London: Lawrence Erlbaum Associates.

Other related books:

- Ackerman, M. S., C. A. Halvorson, T. Erickson and W. A. Kellog (eds) (2008) *Resources, Co-Evolution and Artifacts*, London: Springer.
- Bødker, K., Finn Kensing and Jesper Simonsen (2004) *Participatory IT Design: Designing for Business and Workplace Realities*, Cambridge, MA: MIT Press.
- Dittrich, Y., C. Floyd and R. Klischewski (eds) (2002) *Social Thinking –Software Practice*, Cambridge, MA: MIT Press.
- Grønbaek, Kaj and Randall Trigg (1999) *From Web to Workplace: Designing Open Hypermedia Systems*, Cambridge, MA: MIT Press.
- Simonsen, J., J. O. Bærenholdt, M. Büscher and J. D. Scheuer (eds) (2010) *Design Research: Synergies from Interdisciplinary Perspectives*, London: Routledge.

Special journal issues on Participatory Design:

- CACM: *Communications of the ACM*, 36(4), 1993 (eight articles).
- CoDesign: *International Journal of CoCreation in Design and the Arts*, 8(2–3), 2012 (six articles).
- CSCW: *Computer Supported Cooperative Work: The Journal of Collaborative Computing*, 7(3–4), 1998 (nine articles).
- DI: *Design Issues*, 28(3), 2012 (eight articles).
- HCI: *Human–Computer Interaction*, 11(3), 1996 (five articles).
- SJIS: *Scandinavian Journal of Information Systems*, 21(1), 2009 (three articles).
- SJIS: *Scandinavian Journal of Information Systems*, 22(1), 2010 (one article and debate section).

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